Amlogic

Model Conversion User Guide

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变更记录

Issue 1.6 (2024-07-05)

第七版

1、2.2.3 章节更新 adla_tool 支持的 Framework 版本信息

2、3.4.1 章节添加新的 Platform 信息

3、4.2 章节更新 Quantize optimization tool

4、新增章节 4.4 NNIDE_Tool

5、移除第 5 章节

6、2.2.2 章节增加获取工具的方法

Issue 1.5 (2023-10-20)

第六版

1. 更新 adla_tool 支持的 Framework 版本信息

2. 新增 feature 量化算法以及对应参数说明：

    (1) hist_percentile

    (2) Hist_asymm_percentile

    (3) history_moving-avg

3. 新增其他工具对应的介绍，包括：

    (1) adla_info_tool

    (2) quantize parameter search tool

    (3) NNAPI_Optimize_tool

Issue 1.4 (2023-5-29)

第五版

1. 新增支持 uint8 量化转换

2. 新增支持导出 demo code 功能

3. 新增集成 adla_info 工具

4. 新增支持指定 ONNX、Tflite 模型的输出层

Issue 1.3 (2023-03-20)

第四版

1. 新增 feature 量化算法:

(1) kl_symm_divergence

(2) kl_asymm_divergence

(3) moving-avg、percentie

2. 新增 Feature 量化算法参数:

(1) --quantize-algo、--optimize-algo

(2) --kl-threshold-size、--divergence-nbins

(3) --percentile、--moving-alpha

(4) --neg-scale

3. 新增 Weights 量化算法：mle

4. 新增 Weights 量化算法参数：

(1) --weights-quantize-algo、--weights-optimize-algo

(2) --weights-threshold-size、--weights-neg-scale

Issue 1.2 (2022-11-21)

第三版

1. 更新工具版本对应的 Ubuntu 环境，添加工具包中文档说明

2. 新增参数--dtypes 使用说明

3. 新增 quantized tflite 转换指导

4. FAQ 章节新增模型转换错误和板测精度错误确认步骤

Issue 1.1 (2022-08-30)

第二版

1. 更新了各个 Framework 对应的版本

2. 新增模型转换流程图

3. 添加 int8 perlayer 的转换说明

4. 添加 Paddle 模型的转换命令

Issue 1.0 (2022-05-30)

初始版本.

目录

1. 简介

本文将详细介绍使用 Aala_tool 工具进行模型转换的过程以及细节。指导公司内部或者外部相关开发人员进行模型转换。

当前我们的 Aala_tool 工具支持:caffe、pytorch、onnx、mxnet、darknet、tensorflow、keras、tflite、Paddle 等框架对应的模型。如果开发者使用的是其他框架模型，需先将模型转换成上述支持中的一种。

2. 工具介绍

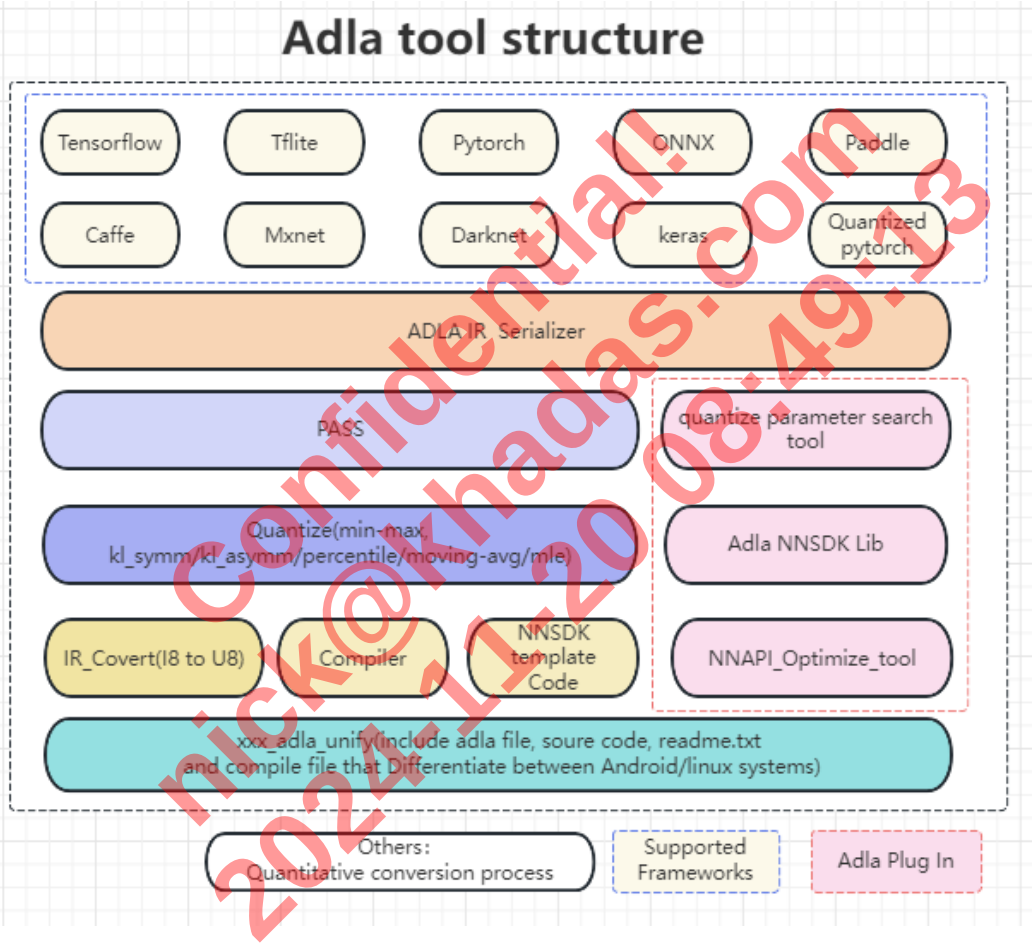
2.1 简介

当前将模型转换出 adla 文件需要的工具:

adla-toolkit 模型工具，用于将模型导入、量化、并转换出最终的 adla 文件。

2.2 模型转换工具

2.2.1 工具结构



2.2.2 工具目录简介

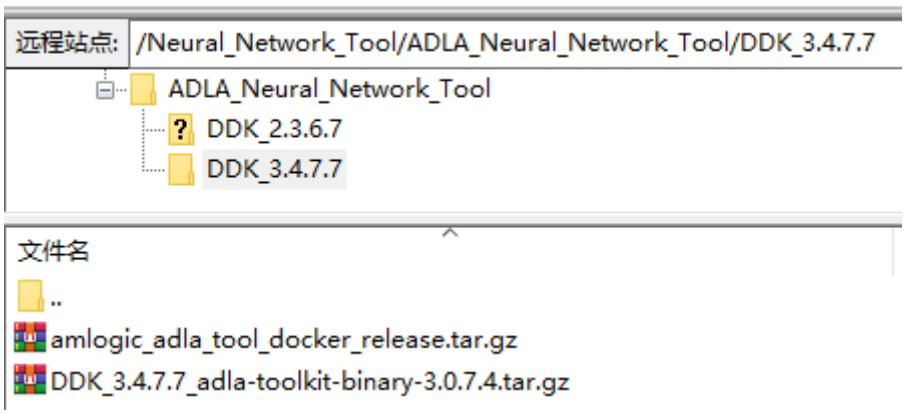
获取工具方式：

FTP server: ftp-china.amlogic.com

account: amlogic_nn_tool_release.guest

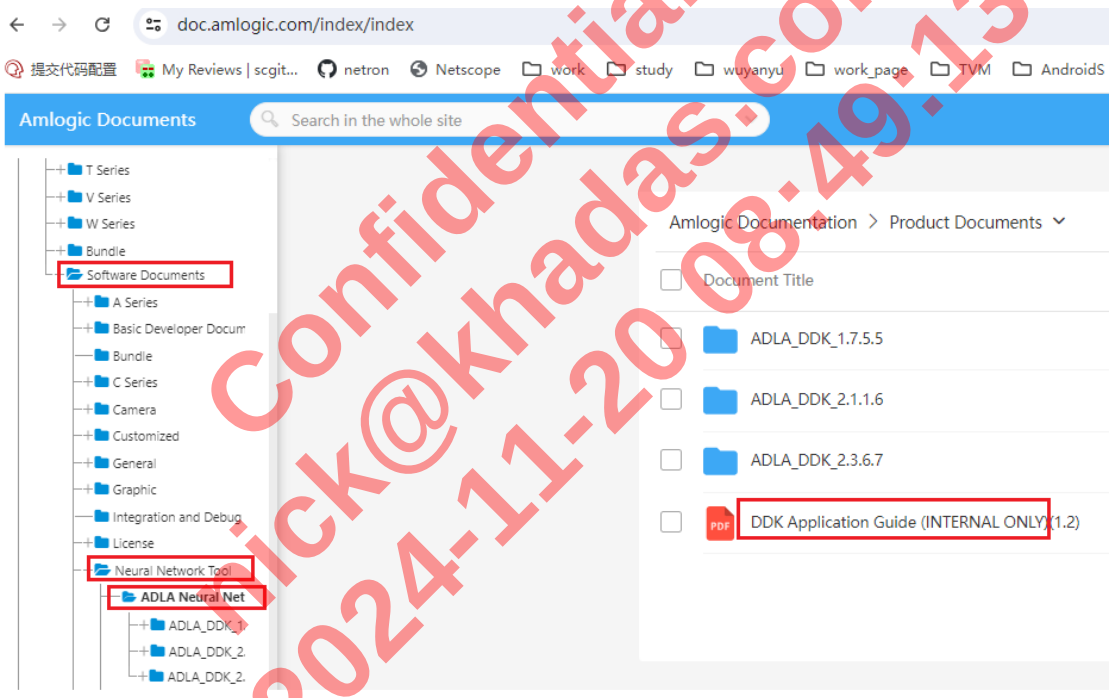
FTP server Path： /Neural_Network_Tool/ADLA_Neural_Network_Tool

获取对应 DDK 版本目录下的工具，如 DDK_3.4.7.7



当前 adla_tool 通过 ftp server release，客户请联系 amlogic 对接人，从申请文档处的 DDK Application Guide (INTERNAL ONLY).doc 中获取最新的访问密码并提供给客户。获取方式如下图：

Link：https://doc.amlogic.com/index/index



Note：在电脑上使用 filezilla 应用程序，使用上面的账号和获取到的密码登录即可获取相应的工具。端口默认或设置为 21。

File In Adla_tool_binary-x.x.x	Description
bin/	Directory containing the executable file for adla tool
adla_convert	soft link to -> ./adlalib/adla_convert

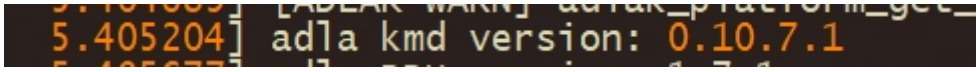
adlalib\	Directory containing the executable file for adla_tool
adla_convert	binary conversion command file,convert model to adla format
adla_plug_in\	Some plugins for adla tools
common\	Compiler and Configuration Files
template\	Template code and nnsdk package
android\	Template code for Android environment
linux\	Template code for Linux environment
example\	The demo code of nnsdk, will be readjusted in the future
nnsdk\	nnsdk development kit
tools\	some auxiliary tools
adla_info_tool\	adlainfo tool,parse adla file information
adla_visualization_tool	adla netron tool, adla file visualization tool
Quantize_optimization_tool	Optimize quantization accuracy tools
tflite_refactorer_tool	tflite refactorer tool, Applicable to use NNAPI to integrate apk scenarios
input_npy_txt_bin_generate.py	Generate random input npy/txt/bin file script
demo/	Directory containing the samples for adla tool
convert_adla.sh	samples model conversion script
data/	Model quantization picture
dataset.txt	Model Quantization Dataset
model_source/	Models corresponding to each framework supported by adla
Readme.txt	Readme file
release_notes.txt	Release notes file

2.2.3 环境依赖

当前 adla_tool 是基于 Ubuntu20.04 python3.8.10 环境下编的 binary tool，客户只需保持基础环境是 Ubuntu20.04 python3.8.10 即可，不需要额外安装其他环境。

板侧 Driver 版本跟工具版本对应关系：

1. 重启板子获取串口内核日志，或者直接执行命令：dmesg |grep version



2.adla 对应的 Driver version 为 0.10.7.1，当前只关注前三位即可。

3.Driver 版本号与 DDK 版本以及 adla_tool 版本的对应关系如下表格

DDK_Version	**Driver Version**	**Adla_tool Version**
DDK 1.2.2.4	0.10.2	adla-toolkit_0.9.3.6/1.0.0.6
DDK 1.4.3.5	0.10.4	adla-toolkit_1.2.0.7
DDK 1.7.5.5	0.10.7	adla-toolkit_1.2.0.9
DDK 2.1.1.6	1.2.0	adla-toolkit_1.5.10 adla-toolkit_1.6.10.3
DDK 2.3.6.7	1.3.0	adla-toolkit_2.0.11.0
DDK 3.4.7.7	1.4.0	adla-toolkit_3.0.7.4

Note：

1、Driver 版本号：第二步中获取到的版本号

2、Adla_tool version: 工具版本号，需与 Driver 版本匹配

3、DDK_Version：只是一个对外 release 的 DDK 版本号

当前支持的 framework 版本以及模型加载方式为：

Framework	Version	Model load(Remark)
Tensorflow	2.15.0	Tf.compat.v1.lite.TFLiteConverter.from_frozen_graph()
Pytorch	2.0.0	torch.jit.load(model_path)
Onnx	1.12.0	onnx.load(model_path)
Mxnet	1.6.0	mx.model.load_checkpoint(prefix_path, prefix_epoch)
Darknet	https://github.com/siju-samuel/darknet	

Caffe	apt-get install caffe-cpu	caffe.Net(proto_file,blob_file,caffe.TEST)
tflite	2.15.0	tflite.Model.GetRootAsModel tflite.Model.Model.GetRootAsModel
Keras	3.0.5	tf.keras.models.load_model(model_path)
Paddle	2.4.2	Paddle.jit.load(model_path)

3. 模型转换

3.1 模型转换简介

模型转换过程是先将各个 framework 模型转换成统一格式的 ADLA IR，然后进行 int8、int16、uint8 量化，再转成基于我们平台能加载并运行的 adla 格式文件。

3.2 量化简介

量化：量化是将以往用 32bit 或者 64bit 表达的浮点数用 16bit、8bit 或者更低的 4bit 方式进行存储。

当前 ADLA 工具量化是复用 tensorflow 进行量化，支持 int8(Perlayer/Perchannel)、int16、uint8 三种量化方式。

3.2.1 量化详解

uint8：将 float32 数据量化成 unsigned int8 的数据。即:将最大/最小范围区间映射到 0~255 区间。

int8：将 float32 数据量化成 signed int8 的数据。即:将最大/最小范围区间映射到-128~127 区间。

Int16：将 float 的数据量化成 signed int16 的数据。即:将最大/最小范围映射到-32768 到 32767 区间。

int8/uint8 量化参数计算公式：(scale:映射比例, zero_point：零点的位置)

scale = (float_max_value - float_min_value)/(quantized_max - quantized-min)

zero_point = quantized_max - float_max_value/scale

Int16 量化参数计算公式：(scale:映射比例)

float_max_temp = max(|float_max_value |, |float_min_value|)

scale = float_max_temp/quantized_max(32767)

zero_point = 0

例如：某个节点输出的最小/最大值范围为（-4~1.076）

uint8 量化参数计算：

scale = (1.076 -(-4))/(255-0) = 0.0199

zero_point = 255 - 1.08/scale = 201

int8 量化参数计算：

scale = (1.076 -(-4))/(127- (-128)) = 0.0199215

zero_point =127 - 1.076/scale = 73

Int16 量化参数计算：

float_max_temp = max(|-4|, |1.08|)

scale =4/32767= 0.000122074

3.2.2 量化方法介绍

模型量化包括 Feature 量化和 Weights 量化。

Feature 量化方法：

1. MinMax 量化（Default）：直接从 fp32 张量中选取最大最小值来确认动态范围

2. kl_symm_divergence：KL 对称量化算法，将 float32 数值分布和 int8 数值分布抽象成两个分布，用阈值 |T| 来更新这两个数值分布，并用 KL 散度来衡量这两个分布的相似性。

3. Kl_asymm_divergence：KL 非对称量化，将 float32 数值分布和 int8 数值分布抽象成两个分布，用[Min,Max] 来更新这两个数值分布，同时使用一个缩放因子 scale 来缩小[Min,Max]范围，并用 KL 散度来衡量这两个分布的相似性。

4. moving-avg：滑动平均最大最小值。

5. Percentile：基于百分比系数来调整 fp32 张量中最大最小值的动态范围。该算法对 Feature 数据分布存在个别极大的离群点数值的模型更为有效。

6. hist_percentile：直方图百分比系数量化算法

7. hist_asymm_percentile：直方图非对称百分比系数量化算法

8. history_moving-avg：历史系数滑动平均量化算法

Weights 量化类型：per-channel 和 per-layer。

Weights Per-layer 量化：conv2d/Dwconv2d 等算子的 filter tensor 量化时统计一个 min/max 并进行量化，per-layer 量化能减小在单板上运行时占用的 DDR 内存和带宽。

Weights Per-channel 量化：conv2d/Dwconv2d 等算子的 filter tensor 量化时，根据 filter 个数分别统计对应的 min/max 值并进行量化，每一个 filter 对应一组量化参数，per-channel 量化能提升量化后的精度。特别对一些 conv 算子的 filter 范围分布差异较大的模型，使用 per-channel 量化，精度提升明显。

Weights 量化方法：当前有 MinMax 和 MLE 量化方法。

1. MinMax 量化（Default）：直接从 fp32 张量中选取最大最小值来确认动态范围

2. MLE:最大似然估计量化方法，对 perlayer、perchannel 量化均有效.

3.2.3 量化方式选取指导

量化方式选取规则：

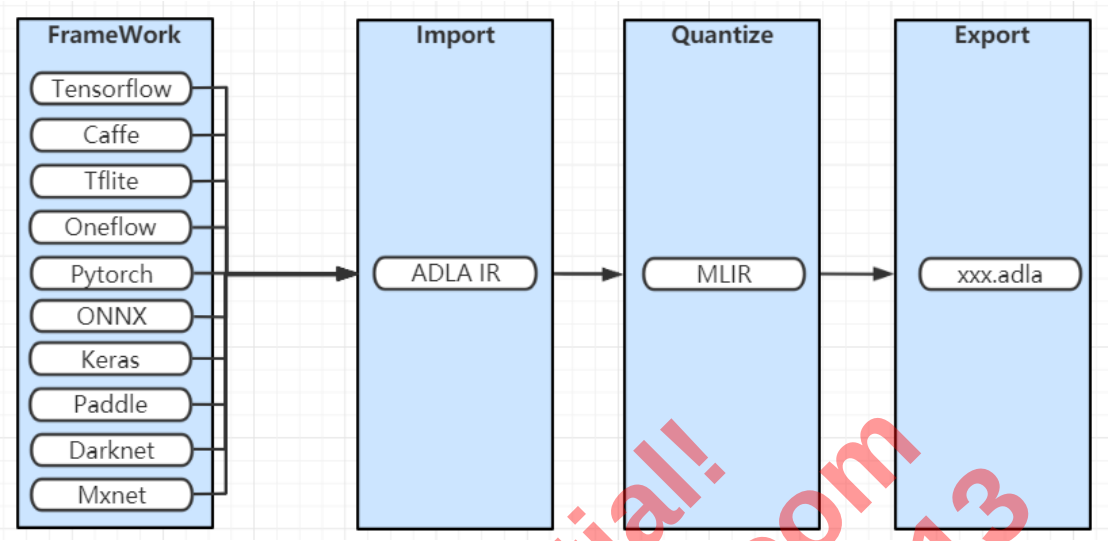
1. 一般量化类型建议使用当前默认值，即：int8 perlayer 量化

2. 如果 int8 perlayer 量化的精度不够，可以设置成 perchannel 量化。或者使用 int16 量化。

3. 当前 feature 和 weights 都默认使用的是 MinMax 量化算法。Feature 算法和 weights 算法在某些参数条件下能提升模型精度，weights 和 feature 量化算法分别平均能提升 0.5%左右的精度。每个模型的最优量化精度对应的参数均不一致，需要客户遍历量化参数，并进行量化后的精度验证来找出最优结果。Feature+weights 算法结合起来对精度的提升效果也需遍历更多参数才能获得。

4. Feature 量化算法选择跟 weights 的 per-layer/perchannel 量化无关

3.3 模型转换流程图



3.4 模型转换

当前模型转换过程都是在 adla-toolkit 目录下进行的。可执行文件均在 bin 目录下。

3.4.1 默认量化算法

1、Feature/weights Min-Max 量化:

Caffe/Darknet：

```
./bin/adla_convert  --model-type        caffe(darknet)  \
        --model                         xx.prototxt(xxx.cfg)            \
        --weights                       xx.caffemodel(xxx.weights)   \
        --quantize-dtype                int8(int16/uint8)  \
        --source-file                   dataset.txt    \
        --channel-mean-value            "128,128,128,128"  \
        --outdir                        convert_output  \
        --target-platform               PRODUCT_PID0XA001 \
         --disable-per-channel          False(default True)   \
        --export-template-code True(default True)   \
         --system-env  'android' \
        #--batch-size                   4(default 1)
```

Onnx/Pytorch：

```
./bin/adla_convert  --model-type        onnx(tflite) \
        --model                         ./xxxx.onnx(xxx.tflite)  \
        --quantize-dtype                int8(int16/uint8)  \
        --inputs                        "0 1 2"  \
        --input-shapes                  "3,288,288#3,288,288#3,288,288"  \
        --source-file                   dataset.txt  \
        --channel-mean-value            "128,128,128,128#0,0,0,255#0,0,0,255"  \
        --outdir                        convert_output  \
        --target-platform               PRODUCT_PID0XA001 \
        --batch-size                    1   \
        --disable-per-channel           False(default True)   \
        --export-template-code  True(default True)   \
         --system-env  'android'  \
```

```
        # --outputs “ouput0 output1”   #onnx/tflite supports specifying output nodes
         #--shape-with-batch          False#False#False  \
        #--dtypes                     “float32#float32#float32”
```

Mxnet:

```
./bin/adla_convert --model-type         mxnet \
        --model                      ./mobilenet0.25-symbol.json \
        --weights                    ./mobilenet0.25-0000.params \
        --quantize-dtype             int8(int16/uint8)  \
        --inputs                     "data" \
        --input-shape                "3,224,224" \
        --channel-mean-value         "0,0,0,255" \
        --source-file                dataset.txt  \
        --outdir                     convert_output  \
        --target-platform            PRODUCT_PID0XA001 \
        --export-template-code  True(default True)   \
        --system-env  ‘android’   \
        #--batch-size                4(default 1)
```

Tensorflow：

```
./bin/adla_convert --model-type         tensorlflow  \
        --model                      xxxx.pb \
        --quantize-dtype             int8(int16/uint8)    \
        --inputs                     data \
        --input-shape                112,112,3  \
        --outputs                     fc1/mul     \
        --channel-mean-value         "0,0,0,255" \
        --source-file                dataset.txt  \
        --outdir                     convert_output  \
        --target-platform            PRODUCT_PID0XA001 \
        --export-template-code    True(default True)   \
        --system-env                 ‘android’
```

Keras:

```
./bin/adla_convert --model-type         keras \
        --model                      xxxx.h5 \
        --inputs                     input1   \
        --quantize-dtype             int8(int16/uint8)    \
        --input-shape                "3,224,224" \
        --channel-mean-value         "0,0,0,255" \
        --source-file                dataset.txt  \
        --outdir                     convert_output  \
        --target-platform            PRODUCT_PID0XA001  \
        --export-template-code  True(default True)   \
        --system-env  ‘android’   \
```

Quantized Tflite:

```
./bin/adla_convert  --model-type        quantized_tflite\
        --model                      resnet18.tflite \
         --target-platform           PRODUCT_PID0XA001  \
          --outdir   test
```

Pytorch:

```
./bin/adla_convert --model-type         pytorch      \
        --model                      xxxx.pt  \
        --quantize-dtype             int8(int16/uint8)  \
        --inputs                     data \
        --input-shape                224,224,3  \
        --channel-mean-value         "0,0,0,255" \
        --source-file                dataset.txt  \
```

```
    --outdir                    convert_output  \
    --target-platform           PRODUCT_PID0XA001 \
    --export-template-code  True(default True)   \
    --system-env  'android'   \
```

Paddle

```
./bin/adla_convert --model-type       paddle      \
    --model                     xxxxx \
    --quantize-dtype            int8(int16/uint8)  \
    --inputs                    data \
    --input-shape               224,224,3  \
    --channel-mean-value        "0,0,0,255" \
    --source-file               dataset.txt  \
    --outdir                    convert_output  \
    --target-platform           PRODUCT_PID0XA001 \
    --export-template-code  True(default True)   \
    --system-env  'android'   \
```

注:

1. --model-type :转换的模型类型，当前支持:onnx、pytorch、caffe、darknet、mxnet、keras、quantized_tflite、tflite、tensorflow、paddle、oneflow 等

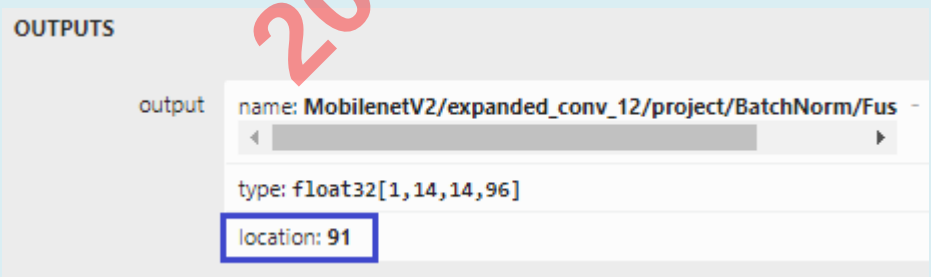
2. --model(模型文件)/--weights(模型权值文件):--model 必须设置参数。model-type 为 caffe/darknet/mxnet 时需要设置--weights 参数

3. --inputs/--input-shapes:在 model-dtype 不为 quantized_tflite/caffe/darknet 时需要设置

4. --inputs:输入节点名称。如果有多个输入，请使用空格分隔每个输入，例如“input1 input2”

5. --outpus:输出节点名称。如果有多个输出，请使用空格分隔每个输出，例如“output1 output2” 当前 outputs 在 model-dtype 为 tensorflow 时必须设置。在 model-type 为 onnx/tflite 时，默认不需要设置，但支持设置中间层为输出节点。

Onnx 模型直接设置中间层的 OUTPUTS 对应的 name 即可。Tflite 模型设置的是中间层 OUTPUTS 对应的 location number。使用 Netron 工具可以查看 onnx 模型的 OUTPUTS name 和 tflite 模型的 OUTPUTS 对应的 location number，tflite locat number 的如下图:



6. --input-shapes:输入节点的尺寸。如果有多个输入，请使用#分隔每个输入的尺寸，例如:“3, 224, 224#3, 299, 299#3, 12, 12”

*7. --shape-with-batch:设置的输入 shape 是否带 batch 信息(**可选项**)。默认值:False。如果设置为 True，则模型实际输入就是 input-shapes 设置的尺寸。如果有多个输入，请用#分隔*

开。例如,模型输入为:”1x224x224x3”,”11x88”时，参数设置为：--input-shapes “224,224,3#11,88” --shape-with-batch “False#True”

8. --dtypes:输入类型，设置输入对应的type 信息(可选项)。默认值:float32。如果有多个输入，请用#分隔开，例如：--dtypes :”float32#float32#int8”

9. --quantize-dtype: 量化类型，当前支持 int8、int16、uint8 三种量化类型

当量化类型为 int8/int16 时,需要设置:--source-file/--channel-mean-value 这组参数。但如果 source-file 是 npy 文件时，则不需要设置 channel-mean-value。

10. --source-file dataset.txt。txt 文件中提供量化图片的路径。当前支持图片和 npy 两种格式。如:./data/cat.jpg 或者 ./data/input_3_224_224.npy

一般建议提供 500~1000 张左右的图片来量化。如果为多输入模型时，dataset.txt 中将一组输入对应的图片放一行并以空格间隔开，如一组输入：./1.jpg ./2.jpg ./3.jpg。如果量化数据有多组时，txt 文件中一行设置一组输入，设置多行即可。npy 文件做量化数据时，shape 需要跟 input-shapes 参数设置的保持一致。

11. --channel-mean-value: 根据训练模型时的预处理方式来设置该预处理参数。参数包括四个值（m1,m2,m3,scale）.前三个值为均值参数，最后一个值为 scale 参数。对于输入为三通道数据（data1，data2，data3），预处理过程为：

Out1 = (data1-m1)/scale

Out2 = (data2-m2)/scale

Out3 = (data3-m3)/scale

例如：如果前处理需将数据归一化到[-1,1]之间，参数设置为：128,128,128,128

如果前处理需将参数归一化到[0,1]之间，参数设置为：0,0,0,255

如果是单通道模型，参数设置方式为：m1,0,0,scale

如果是多输入模型，可以针对多个输入设置不同的 channel-mean-value，以#号隔开，例如：--channel-mean-value "0,0,0,1#127.5,127.5,127.5,127.5#0,0,0,255"

如果图片三通道对应的 scale 均不相同或者模型是非图片输入时，建议先使用 Python 读取模型的输入数据并做完减 mean 除以 scale 的操作后保存成 npy 文件，在模型转换时，量化的 source-file 中提供 npy 文件路径即可

12. --batch-size:设置转换出 adla 文件的 batch-size，当前默认值为 1

13. --iterations:可选参数，默认值 1000，如果 dataset.txt 中的提供的是多组数据，且需要使用所有提供的数据进行量化时，请设置--iterations 参数，确保 iterations*batch-size=多组数据个数

14. --disable-per-channel: 是否禁用逐通道量化方式，默认为 True。设置 True 时，为 perlayer 量化，否则为 perchannel 量化.

15. --outdir:转换出的 adla 文件输出路径，默认值为当前路径下的“xxx_adla_unify”目录

16. --target-platform：指定转换目标平台的 adla 文件。当前默认值：PRODUCT_PID0XA001

当前有如下有效值：

PRODUCT_PID0XA001、PRODUCT_PID0XA002

PRODUCT_PID0XA003、PRODUCT_PID0XA004

PRODUCT_PID0XA005

可以使用如下方式来确认应该设置的值，执行：

cat /proc/cpuinfo 或 cat /proc/cpu_chipid

*查看倒第二行数 Serial 对应的数列码**前四个字符（Serial 后面粗体字部分）***

序列号和参数 PRODUCT_PID0X？ 的对应关系如下图：

PID	Serial
0XA001	Serial: **3d0a**010100000000c613492931563343
0XA002	Serial: **3e0a**0202000000007c404b6c31563553
0XA003	Serial: **360c**0103000000025f904ab31563541
0XA004	Serial: **420a**010100000000daff6e4931583354
0XA005	Serial: **480a**020100000000682763293156365 3

例如：

在串口或者 cmd 窗口执行命令：cat /proc/cpuinfo 后，Serial 如下图

```
Serial          : 360c0103000000009 46bd13930563754
Hardware        : Amlogic
```

此时，参数应该设置为：--target-platform PRODUCT_PID0XA003

Note：

1. *如果没有获取到对应上表格中的序列号，可以执行：cat /proc/cpu_chipid*
2. *如果序列号的前四位没有匹配上，可以只匹配前三位*

17. --export-template-code:是否导出 demo code，默认为 True。设置为 False 时，只转换出 adla 文件

18. --system-env：指定单板系统环境，默认为 linux。跟--export-template-code 配套使用。使用默认参数时，导出的 demo code 中带有 linxu 对应的编译脚本。设置为”android”时，导出的 demo code 中带有 Android 对应的编译脚本。 当前支持[linux,android]俩个有效值，用户根据当前单板环境来设置。然后参考生成的 demo code 中的 Readme.txt 的指导进行编译即可。

结果：最终生成带有 adla 文件的 demo code，如下图：

```
resnet34-v2-7_adla_unify/
├── CMakeLists.txt
├── config.h
├── main.c
├── Readme.txt
└── resnet34-v2-7_int8.adla
```

3.4.2 其他量化算法

一般使用上一章节中的默认量化算法即可。如果默认量化算法的精度不满足需要，可以尝试使用进阶量化算法。

当前适配了多种量化算法，不同量化算法对应的参数各不相同，下面会列出不同量化算法的使用方法。客户如果需要使用进阶量化算法，建议客户参照 4.2 章节中介绍的 Quantize_optimization_tool 工具来使用进阶量化算法。

2、Feature kl_symm_divergence+weights Min-max 量化

以 caffe model 为例：

```
./bin/adla_convert  --model-type        caffe(darknet)  \
        --model                         xx.prototxt(xxx.cfg)            \
        --weights                       xx.caffemodel(xxx.weights)    \
        --quantize-dtype                int8(int16/uint8)  \
        --source-file                   dataset.txt     \
        --channel-mean-value            "128,128,128,128"  \
        --outdir                        convert_output  \
        --target-platform               PRODUCT_PID0XA001 \
        --disable-per-channel           False   \
        --quantize-algo                 “kl_symm_divergence”  \
        --optimize-algo                 “kl-dis”  \
        --kl-threshold-size             8192\
        --divergence-nbins              16384 \
        --export-template-code  True(default True)   \
        --system-env  ‘android’
```

注:

1. *跟默认的 MinMax 量化相比，KL 对称量化新增了四个参数，其他参数的设置跟 MinMax 量化时一致*
2. *--quantize-algo：默认值为:"min-max",当前设置量化算法类型为“kl_symm_divergence”。*
3. *--optimize-algo：度量 float32 数值分布和 int8 数值分布的相似性的方法。默认值:“kl-dis”,设置 kl_symm_divergence 时，对应的有效值为:"kl-dis"/"mse-dis"*
4. *--kl-threshold-size: KL 量化时 Feature 是否使用 kl 量化的阈值。如果量化的 Feature size 过小，不宜使用 KL 量化。默认值：2048，一般建议直接使用默认值。*
5. *--divergence-nbins:KL 量化时，直方图中 bin 的个数。默认值：2048，建议取值:2048/4096/8192/12288/16384*

部分模型使用 kl 对称量化算法，使用不同参数进行量化后的精度如下图：（第一行结果为 min-max 量化对应精度）

quantize_algo	optimize_algo	num_histogram_bins	kl_thread_size	resnet_18		resnet_34		googlenet-7		shufflenet-v2-10	
				TOP1	TOP5	TOP1	TOP5	TOP1	TOP5	TOP1	TOP5
NA	NA	NA	NA	62.72%	84.64%	67.24%	88.22%	67.50%	88.20%	49.98%	72.98%
kl_symm_divergence	kl-dis	2048	2048	63.22%	84.66%	67.46%	88.24%	67.36%	88.44%	46.06%	70.60%
			4096	63.26%	84.54%	67.82%	88.28%	67.64%	88.20%	46.14%	70.72%
			8192	62.60%	84.72%	67.12%	88.00%	67.56%	88.40%	46.36%	70.82%
			12288	62.60%	84.72%	67.12%	88.00%	67.28%	88.36%	48.62%	72.08%
			16384	62.60%	84.72%	67.12%	88.00%	67.34%	88.36%	48.62%	72.08%
	mse-dis	2048	2048	62.92%	84.74%	67.18%	87.96%	67.22%	88.38%	45.38%	69.50%
			4096	62.38%	84.70%	66.78%	87.88%	67.20%	88.36%	50.30%	72.86%
			8192	62.38%	84.70%	66.78%	87.88%	67.12%	88.34%	50.30%	72.86%
			12288	62.38%	84.70%	66.78%	87.88%	66.96%	88.42%	49.84%	72.88%
			16384	62.38%	84.70%	66.78%	87.88%	67.20%	88.32%	49.84%	72.88%
		4096	2048	62.52%	84.72%	67.14%	88.00%	67.16%	88.28%	50.30%	72.86%
			4096	62.52%	84.72%	67.14%	88.00%	67.40%	88.32%	50.30%	72.86%
			8192	62.52%	84.72%	67.14%	88.00%	67.22%	88.24%	50.30%	72.86%
			12288	62.52%	84.72%	67.14%	88.00%	67.20%	88.16%	49.84%	72.88%
			16384	62.52%	84.72%	67.14%	88.00%	67.26%	88.28%	49.84%	72.88%
		8192	2048	62.60%	84.76%	67.10%	88.12%	67.34%	88.32%	49.66%	72.64%
			4096	62.60%	84.76%	67.10%	88.12%	67.16%	88.12%	49.66%	72.64%
			8192	62.60%	84.76%	67.10%	88.12%	67.42%	88.20%	49.66%	72.64%
			12288	62.60%	84.76%	67.10%	88.12%	67.28%	88.06%	49.86%	73.04%
			16384	62.60%	84.76%	67.10%	88.12%	67.34%	88.18%	49.86%	73.04%
		12288	2048	62.86%	84.74%	67.12%	87.92%	67.40%	88.20%	50.16%	72.66%
			4096	62.86%	84.74%	67.12%	87.92%	67.48%	88.24%	50.16%	72.66%
			8192	62.86%	84.74%	67.12%	87.92%	67.26%	88.30%	50.16%	72.66%
			12288	62.86%	84.74%	67.12%	87.92%	67.38%	88.22%	49.94%	73.00%
			16384	62.86%	84.74%	67.12%	87.92%	67.40%	88.34%	49.94%	73.00%
		16384	2048	62.68%	84.56%	67.00%	88.08%	67.06%	88.16%	49.94%	72.86%
			4096	62.68%	84.56%	67.00%	88.08%	67.42%	88.36%	49.94%	72.86%
			8192	62.68%	84.56%	67.00%	88.08%	67.14%	88.06%	49.94%	72.86%
			12288	62.68%	84.56%	67.00%	88.08%	67.10%	88.22%	50.14%	72.96%
			16384	62.68%	84.56%	67.00%	88.08%	67.46%	88.30%	50.14%	72.96%

3、Feature Kl_asymm_divergence+weights Min-max 量化

以 caffe model 为例：

```
./bin/adla_convert  --model-type        caffe(darknet)  \
        --model                         xx.prototxt(xxx.cfg)            \
        --weights                       xx.caffemodel(xxx.weights)    \
        --quantize-dtype                int8(int16/uint8)  \
        --source-file                   dataset.txt     \
        --channel-mean-value            "128,128,128,128"  \
        --outdir                        convert_output  \
        --target-platform               PRODUCT_PID0XA001 \
        --disable-per-channel           False   \
        --quantize-algo                 “kl_asymm_divergence”  \
        --optimize-algo                 “kl-dis”  \
        --kl-threshold-size             2048  \
        --divergence-nbins              2048   \
        --neg-scale                     0.9 \
        --export-template-code  True(default True)   \
        --system-env  ‘android’
```


注:

1. *跟默认的 MinMax 量化相比，KL 对称量化新增了四个参数，其他参数的设置跟 MinMax 量化时一致*
2. *--quantize-algo：默认值为:"min-max",当前设置量化算法类型为“kl_symm_divergence”。*
3. *--optimize-algo：度量 float32 数值分布和 int8 数值分布的相似性的方法。默认值:“kl-dis”,设置 kl_symm_divergence 时，对应的有效值为:"kl-dis"/"js-dis"/"cos-dis"*
4. *--kl-threshold-size: KL 量化时 Feature 是否使用 kl 量化的阈值。如果量化的 Feature size 过小，不宜使用 KL 量化。默认值：2048，一般建议直接使用默认值。*

5. --divergence-nbins:KL 量化时，直方图中 bin 的个数。默认值：2048，建议取值:2048/4096/8192/12288/16384

6. --neg-scale: 默认值：1.0

部分模型使用 Feature 的 percentile 算法量化后精度结果如下图：(首行数据为 feature min-max 精度)

quantize_algo	optimize_algo	num_histogram_bins	kl_thread_size	neg_scale	Number of pictures for	resnet_18		shufflenet-v2-10		squeezenet1.1-7		mnasnet1_0		Darknet19	
						TOP1	TOP5	TOP1	TOP5	TOP1	TOP5	TOP1	TOP5	TOP1	TOP5
NA	NA	NA	NA	NA	200	62.72%	84.64%	49.98%	72.98%	50.14%	74.92%	48.66%	73.16%	72.92%	90.92%
kl_asymm_divergence	cos-dis	2048	2048	0	200	62.98%	84.74%	49.56%	73.02%	50.40%	74.98%	48.58%	73.20%	73.24%	91.38%
				0.001	200	63.08%	84.96%	49.68%	73.04%	50.06%	74.50%	48.74%	73.32%	73.28%	91.20%
				0.01	200	62.92%	84.66%	49.44%	72.96%	50.06%	74.50%	48.66%	73.20%	73.26%	91.30%
				0.1	200	62.96%	84.70%	49.92%	73.20%	50.06%	74.50%	48.58%	73.24%	73.26%	91.06%
				1	200	63.08%	84.96%	50.00%	72.82%	50.06%	74.50%	48.94%	73.12%	73.66%	91.40%
				1.2	200	62.96%	84.58%	49.82%	73.10%	50.06%	74.50%	48.66%	73.12%	73.42%	91.48%
				1.5	200	63.08%	84.96%	49.58%	72.86%	50.06%	74.50%	48.88%	73.30%	73.50%	91.46%
			4096	0	200	62.98%	84.74%	49.78%	73.14%	50.54%	74.84%	48.84%	73.16%	73.30%	91.30%
				0.001	200	62.84%	84.76%	49.84%	72.98%	50.06%	74.50%	48.66%	73.04%	73.50%	91.24%
				0.01	200	63.08%	84.96%	49.92%	72.94%	50.06%	74.50%	48.78%	73.34%	73.22%	91.36%
				0.1	200	63.08%	84.96%	49.82%	72.98%	50.06%	74.50%	49.14%	73.16%	73.40%	91.14%
				1	200	63.10%	84.66%	49.62%	73.20%	50.06%	74.50%	48.88%	73.16%	73.34%	91.24%
				1.2	200	63.08%	84.96%	49.82%	72.94%	50.06%	74.50%	49.10%	73.34%	73.40%	91.34%
				1.5	200	63.08%	84.96%	50.00%	73.14%	50.06%	74.50%	48.88%	73.34%	73.34%	91.36%
			8192	0	200	62.96%	84.64%	50.14%	72.96%	50.72%	75.08%	48.64%	73.32%	73.36%	91.20%
				0.001	200	63.02%	84.78%	50.18%	72.88%	50.06%	74.50%	48.80%	73.32%	73.28%	91.36%
				0.01	200	63.08%	84.96%	49.90%	73.12%	50.06%	74.50%	48.76%	73.24%	73.28%	91.30%
				0.1	200	63.08%	84.96%	49.88%	72.88%	50.06%	74.50%	48.52%	73.22%	73.42%	91.06%
				1	200	63.08%	84.96%	49.90%	72.92%	50.06%	74.50%	48.94%	73.38%	73.16%	91.30%
				1.2	200	63.34%	84.80%	50.12%	72.98%	50.06%	74.50%	48.90%	73.28%	73.40%	91.26%
				1.5	200	63.08%	84.96%	49.66%	73.02%	50.06%	74.50%	48.68%	73.38%	73.36%	91.28%

4、Feature Percentile+weights Min-max 量化

以 caffe model 为例：

```
./bin/adla_convert  --model-type        caffe(darknet)  \
        --model                         xx.prototxt(xxx.cfg)        \
        --weights                       xx.caffemodel(xxx.weights)   \
        --quantize-dtype                int8(int16/uint8)  \
        --source-file                   dataset.txt     \
        --channel-mean-value            "128,128,128,128"  \
        --outdir                        convert_output  \
        --target-platform               PRODUCT_PID0XA001 \
        --disable-per-channel           False   \
        --quantize-algo                 "percentile"  \
        --percentile                    0.9999 \
        --export-template-code  True(default True)  \
        --system-env  'android'
#       --asymm-percentile      0.999
```


注:

1. 跟默认的 MinMax 量化相比，Percentile 量化新增了两个参数，其他参数的设置跟 MinMax 量化时一致

2. --quantize-algo：默认值为:"min-max",当前设置量化算法类型为“percentile”。

*3. --percentile：百分比系数，默认值：0.9999，建议取值范围：**0.99 ~ 1.0***

4. --asymm-percentile: 非对称百分比系数，默认值：-1.0，建议取值范围：0.99 ~ 1.0. 当为默认值时，为对称百分比量化算法。当设置为其他值时，为非对称百分比量化算法。

部分模型使用 Feature 的 percentile 算法量化后精度结果如下图：(首行数据为 feature min-max 精度)

quantize_algo	optimize_algo	num_histogram_bins	kl_thread_size	mber of pictures for Quant	resnet_18		resnet_34		googlenet-7		shufflenet-v2-10		squeezenet1.1-7	
					TOP1	TOP5	TOP1	TOP5	TOP1	TOP5	TOP1	TOP5	TOP1	TOP5
NA	NA	NA	NA	100	62.72%	84.64%	67.24%	88.22%	67.50%	88.20%	49.98%	72.98%	50.14%	74.92%
percentile	NA	NA	0.9991	100	62.84%	84.62%	66.92%	87.88%	67.02%	88.38%	48.86%	72.62%	50.40%	74.80%
	NA	NA	0.9992	100	62.98%	84.68%	66.76%	87.90%	67.08%	88.28%	48.72%	72.60%	50.22%	74.64%
	NA	NA	0.9993	100	62.80%	84.74%	66.88%	88.12%	67.18%	88.38%	48.74%	72.64%	50.42%	74.70%
	NA	NA	0.9994	100	62.92%	84.70%	66.98%	87.96%	67.16%	88.30%	49.44%	72.66%	50.60%	74.66%
	NA	NA	0.9995	100	63.16%	84.72%	67.14%	87.94%	67.26%	88.28%	49.52%	73.06%	50.18%	75.06%
	NA	NA	0.9996	100	63.04%	84.62%	67.10%	88.10%	67.22%	88.50%	49.40%	72.60%	50.50%	74.78%
	NA	NA	0.9997	100	63.20%	84.64%	67.18%	88.06%	67.24%	88.40%	49.86%	73.14%	50.38%	74.32%
	NA	NA	0.9998	100	63.06%	84.66%	66.98%	88.06%	67.62%	88.24%	49.58%	72.84%	50.12%	74.50%
	NA	NA	0.9999	100	62.88%	84.72%	66.96%	88.36%	67.48%	88.32%	49.82%	72.98%	50.04%	74.68%

5、Feature Moving-avg+weights Min-max 量化

以 caffe model 为例：

```
./bin/adla_convert  --model-type        caffe(darknet)  \
        --model                         xx.prototxt(xxx.cfg)          \
        --weights                       xx.caffemodel(xxx.weights)    \
        --quantize-dtype                int8(int16/uint8)  \
        --source-file                   dataset.txt     \
        --channel-mean-value            "128,128,128,128" \
        --outdir                        convert_output  \
        --target-platform               PRODUCT_PID0XA001 \
        --disable-per-channel           False   \
        --quantize-algo                 "moving-avg" \
        --moving-alpha                  0.2  \
        --export-template-code  True(default True)   \
        --system-env  'android'
```

注:

1. 跟默认的 MinMax 量化相比，moving-avg 量化新增了两个参数，其他参数的设置跟 MinMax 量化时一致

2. --quantize-algo：默认值为:"min-max",当前设置量化算法类型为“moving-avg”。

*3. --moving-alpha：moving-alpha 系数，默认值：0.2，建议取值范围：**0.2~0.9***

部分模型使用 Feature 的 moving-avg 算法量化后精度结果如下图：(首行数据为 feature min-max 精度)

quantize_algo	optimize_algo	num_histogram_bins	kl_thread_size	mber of pictures for Quant	resnet_18		resnet_34		googlenet-7		shufflenet-v2-10		squeezenet1.1-7	
					TOP1	TOP5	TOP1	TOP5	TOP1	TOP5	TOP1	TOP5	TOP1	TOP5
NA	NA	NA	NA	100	62.72%	84.64%	67.24%	88.22%	67.50%	88.20%	49.98%	72.98%	50.14%	74.92%
moving-avg	NA	0.2	NA	100	60.46%	84.56%	64.80%	88.18%	67.08%	88.30%	49.08%	73.18%	50.34%	74.34%
	NA	0.3	NA	100	61.30%	84.70%	65.54%	88.28%	67.10%	88.42%	49.60%	72.64%	50.22%	74.44%
	NA	0.4	NA	100	61.42%	84.50%	66.00%	88.12%	67.06%	88.26%	49.66%	73.00%	50.78%	74.28%
	NA	0.5	NA	100	61.78%	84.78%	66.08%	87.98%	67.28%	88.44%	49.78%	72.82%	49.94%	73.82%
	NA	0.6	NA	100	62.16%	84.54%	66.46%	87.96%	67.46%	88.32%	49.76%	73.12%	50.06%	74.76%
	NA	0.7	NA	100	62.42%	84.70%	66.82%	87.98%	67.40%	88.42%	49.98%	72.72%	50.42%	74.44%
	NA	0.8	NA	100	62.80%	84.62%	66.56%	87.98%	67.32%	88.52%	50.28%	72.98%	50.00%	74.26%
	NA	0.9	NA	100	62.86%	84.54%	67.00%	88.00%	67.32%	88.30%	49.72%	73.36%	49.98%	74.82%

6、Feature history_moving-avg + weights Min-max 量化

以 caffe model 为例：

```
./bin/adla_convert  --model-type        caffe(darknet)  \
        --model                         xx.prototxt(xxx.cfg)          \
        --weights                       xx.caffemodel(xxx.weights)    \
        --quantize-dtype                int8(int16/uint8)  \
        --source-file                   dataset.txt     \
        --channel-mean-value            "128,128,128,128" \
        --outdir                        convert_output  \
        --target-platform               PRODUCT_PID0XA001 \
        --disable-per-channel           False   \
        --quantize-algo                 "history_moving-avg" \
        --moving-alpha                  0.2  \
```

```
        --export-template-code  True(default True)   \
        --system-env  ‘android’
```

注:

1. *history_moving-avg 的参数跟 moving-avg 算法的参数一致*
2. *--quantize-algo：默认值为:"min-max",当前设置量化算法类型为“history_moving-avg”。*

7、Feature hist_percentile/hist_asymm_percentile + weights Min-max 量化

以 caffe model 为例：

```
./bin/adla_convert  --model-type        caffe(darknet)  \
        --model                         xx.prototxt(xxx.cfg)          \
        --weights                       xx.caffemodel(xxx.weights)   \
        --quantize-dtype                int8(int16/uint8)  \
        --source-file                   dataset.txt     \
        --channel-mean-value            "128,128,128,128"  \
        --outdir                        convert_output  \
        --target-platform               PRODUCT_PID0XA001 \
        --disable-per-channel           False   \
        --quantize-algo                 "hist_percentile"  \ # ”hist_asymm_percentile”
        --percentile                    0.9999 \
        --export-template-code          True(default True)   \
        --system-env                    ‘android’
```

注:

1. *hist_percentile/hist_asymm_percentile 的参数一致，只需要设置百分比系数--percentile 即可*

2. --quantize-algo 当前设置量化算法类型为"hist_percentile" 或者 ”hist_asymm_percentile”

8、Feature Min-Max+Weights MLE 量化

以 caffe model 为例：

```
./bin/adla_convert  --model-type        caffe(darknet)  \
        --model                         xx.prototxt(xxx.cfg)          \
        --weights                       xx.caffemodel(xxx.weights)   \
        --quantize-dtype                int8(int16/uint8)  \
        --source-file                   dataset.txt     \
        --channel-mean-value            "128,128,128,128"  \
        --outdir                        convert_output  \
        --target-platform               PRODUCT_PID0XA001 \
        --disable-per-channel           True   \
        --weights-quantize-algo         "mle"  \
        --weights-optimize-algo         "cos-dis"  \
        --weights-threshold-size        2048     \
        --weights-neg-scale             0.999  \
        --export-template-code          True(default True)   \
        --system-env                    ‘android’
```

注:

1. *跟 weights 默认的 MinMax 量化相比，mle 量化新增了四个参数，其他参数的设置跟 MinMax 量化时一致*
2. *--weights-quantize-algo：默认值为:"min-max",当前设置量化算法类型为"mle"。*

3. --weights-optimize-algo：默认值(cos-dis),当前有效值为："cos-dis"/"kl-dis"/"js-dis"

4. --weights-neg-scale: weights 量化时的缩放系数，默认值(0.999),当前可设置范围:0.99~1。

部分模型使用 Weights 的 MLE 算法量化后精度结果如下图：(首行数据为 min-max 精度)

Weights quantize algo	optimize algo	weights thread_size	Number of pictures for Quantize	wegiths neg_scale	resnet18-v1-7		resnet34-v2-7		resnet50-v2-7		shufflenet-v2-10		squeezenet1.1-7	
					TOP1	TOP5	TOP1	TOP5	TOP1	TOP5	TOP1	TOP5	TOP1	TOP5
Min-Max	NA	NA	200	NA	64.88%	87.48%	70.94%	89.94%	71.54%	89.90%	49.94%	72.50%	49.98%	74.1%
MLE	cos-dis	2048	200	0.99	64.70%	87.52%	70.88%	90.10%	71.62%	90.18%	51.54%	74.42%	50.60%	74.70%
				0.991	64.98%	87.26%	71.28%	89.90%	71.60%	90.30%	51.36%	74.14%	49.44%	74.58%
				0.999	65.22%	87.44%	70.84%	89.94%	72.12%	90.12%	49.92%	72.98%	49.32%	74.18%
		16384	200	0.99	64.88%	87.48%	71.12%	90.12%	71.54%	89.90%	51.60%	73.98%	49.98%	74.10%
	js-dis	2048	200	0.99	64.88%	87.48%	70.94%	89.94%	71.54%	89.90%	49.94%	72.50%	49.98%	74.10%
		16384	200	0.99	64.88%	87.48%	70.94%	89.94%	71.54%	89.90%	51.96%	74.08%	49.98%	74.10%
	kl-dis	2048	200	0.99	64.88%	87.48%	70.94%	89.94%	71.54%	89.90%	49.94%	72.50%	49.98%	74.10%
		16384	200	0.99	64.88%	87.48%	70.94%	89.94%	71.54%	89.90%	49.94%	72.50%	49.98%	74.10%

9、Feature kl_asymm_divergence + Weights MLE 量化

以 caffe/darknet model 为例：

```
./bin/adla_convert  --model-type        caffe(darknet)  \
        --model                         xx.prototxt(xxx.cfg)            \
        --weights                       xx.caffemodel(xxx.weights)      \
        --quantize-dtype                int8  \
        --source-file                   dataset.txt     \
        --channel-mean-value            "128,128,128,128" \
        --outdir                        convert_output  \
        --target-platform               PRODUCT_PID0XA001 \
       --quantize-algo                  kl_asymm_divergence \
        --optimize-algo                 cos-dis  \
       --kl-threshold-size              4096 \
       --divergence-nbins               4096 \
       --neg-scale                      0.9 \
       --disable-per-channel            True  \
        --weights-quantize-algo         "mle"  \
        --weights-optimize-algo         "cos-dis"  \
        --weights-threshold-size  2048     \
        --weights-neg-scale  0.999  \
        --export-template-code  True(default True)  \
        --system-env  'android'
```


3.5 参数说明

3.5.1 简介

3.4.1 章节中介绍的模型转换过程中的参数是最简洁的使用参数，一般情况下只使用上述参数即可正常转换出 adla 文件。本章节 adla_convert 转换命令的所有参数进行扩展说明，以供使用。

3.5.2 adla_convert 参数解析

```
usage: adla_convert.py [-h] --model-type MODEL_TYPE --model MODEL
                       [--weights WEIGHTS] [--inputs INPUTS]
                       [--outputs OUTPUTS] [--input-shapes INPUT_SHAPES]
                       [--shape-with-batch SHAPE_WITH_BATCH]
                       [--dtypes DTYPES]
                       [--batch-size BATCH_SIZE] [--iterations ITERATIONS]
                       [--outdir OUTDIR] [--quantize-dtype QUANTIZE_DTYPE]
                       [--source-file SOURCE_FILE]
                       [--channel-mean-value CHANNEL_MEAN_VALUE]
                       [--disable-per-channel DISABLE_PER_CHANNEL]
                       [--inference-input-type INFERENCE_INPUT_TYPE]
```

```
[--inference-output-type INFERENCE_OUTPUT_TYPE]
[--target-platform TARGET_PLATFORM]
[--quantize-algo QUANTIZE_ALGO]
[--optimize-algo OPTIMIZE_ALGO]
[--kl-threshold-size KL_THRESHOLD_SIZE]
[--divergence-nbins DIVERGENCE_NBINS]
[--percentile PERCENTILE]
[--asymm-percentile ASYMM_PERCENTILE]
[--moving-alpha MOVING_ALPHA]
[--neg-scale NEG_SCALE]
[--weights-quantize-algo WEIGHTS_QUANTIZE_ALGO]
[--weights-optimize-algo WEIGHTS_OPTIMIZE_ALGO]
[--weights-threshold-size WEIGHTS_THRESHOLD_SIZE]
[--weights-neg-scale WEIGHTS_NEG_SCALE]
[--export-template-code EXPORT_TEMPLATE_CODE]
[--system-env SYSTEM_ENV]
```

Argument	Description
--model-dtype(required)	Requires a string value to denote the type of the convert model. Ranges: caffe, onnx, keras, darknet, tensorflow, quantized_tflite, tflite, pytorch, onnx.
--model(required)	Requires a string value to denote the file path of the imported model file, such as xxx.prototxt, xxx.cfg, xxx.onnx
--weights	Requires a string value to denote the file path of the imported weights file, such as xxx.caffemodel, xxx.weights, xxx..params.
--inputs	Multiple points are separated with spaces. For example,” input1 input2”
--input-shapes	Tensor shape sizes of the same input points are separated with commas. For example, '3,224,224', which represents the tensor shape sizes of one input point. Sizes between different input points are separated with hashtags. For example, '3,224,224#3,299,299#12,1', tensor shape sizes of three input points.
--shape-with-batch	Describes whether the --input-shapes contains the highest batch dimension or not. (**Optional**) None **(Default)** means --input-size-list should be given without batch. Enclose the bool values in quotation marks, and use # to separate the bool values. Take a graph with three input layer as an example, the three input layers with shape “?x224x224x3”, “11x88”, “10”, input-shapes and shape with batch could be given as: --input-shapes “224,224,3#11,88#10” --shape-with-batch “False#True#True”
--dtypes	Describes the input types (**Optional**) float32**(Default)** Enclose the string values in quotation marks, and use # to separate the bool values. Effective value**['float32','int64','int32','bool','uint8']** Take a graph with three input layer as an example, the three input layers with shape “?x224x224x3”, “11x88”, “10”, input-shapes and shape with batch could be given as:

	--input-shapes “224,224,3#11,88#10” --shape-with-batch “False#True#True” --dtypes ‘float32#int32#int64’
--outputs	Multiple points are separated with spaces. For example, 'output1 output2'. **Note:** The current outputs must be set when the model-dtype is tensorflow. When the model-type is onnx/tflite, it does not need to be set by default, but it supports setting the middle layer as an output node.. The Onnx model can directly set the name corresponding to the OUTPUTS of the middle layer. The Tflite model sets the location number corresponding to the middle layer OUTPUTS.
--batch-size	Batch size value. Default value:**1**
--iterations	Number of iterations to run, integer value. (Optional) **1000 (Default)**
--outdir	Requires a string value to denote the output path of the generated adla file, default: “`xxx_adla_unify`” (current folder)
--quantize-dtype	Data type used for quantized data. (Optional) Valid **dtype** values are: **int8:** signed 8bit, default per-channel, if per-layer quantization is required, use it with disable-per-channel parameter **int16:** signed 16bit, default per-channel, if pe-rlayer quantization is required, use it with disable-per-channel parameter **uint8**：unsigned 8bit
--source-file	Dataset path and filename. The file extension indicates the dataset: xxx.txt, set the **image path** or **npy file path** in txt.
--channel-mean-value	Input channels mean value. This parameter should always be given according to running models. **Note:** This parameter is only valid for **int8/int16** quantization and the source-file is **image path**.
--disable-per-channel	default True
--inference-input-type	After converting to adla, the input data type of the model. **Default:** same as quantize-dtype For **int8** quantize, inference input type only supports **float32**, **int8**, **uint8**. For **int16** quantize, inference input type only supports **float32**, **int16**.
--inference-output-type	After converting to adla, the output data type of the model. **Default:** same as quantize-dtype For **int8** quantize, inference input type only supports **float32**, **int8**, **uint8**. For **int16** quantize, inference input type only supports **float32**, **int16**.
--target-platform	Specify the config file to export adla file on the target platform. valid values as below: PRODUCT_PID0XA001**(default)** PRODUCT_PID0XA002 PRODUCT_PID0XA003 PRODUCT_PID0XA004
--quantize-algo	1. **"kl-asymm_divergence":** The kl asymmetric quantization algorithm, If this algorithm is selected, you can set the

	parameters related to the algorithm:**--optimize-algo,--kl-threshold-size ,--divergence-nbins,--neg-scale** 2. **"kl-symm_divergence":** The kl symmetric quantization algorithm, If this algorithm is selected, you can set the parameters related to the algorithm:**--optimize-algo,--kl-threshold-size ,--divergence-nbins** 3. **"percentile":** The percentile algorithm,a percentile-based activation clamping quantification method. If this algorithm is selected, you can set the parameters related to the algorithm: **--percentile** 4. **"moving-avg":** The moving-avg algorithm. If this algorithm is selected. You can set the parameters related to the algorithm **--moving-alpha**
--optimize-algo	Quantitative methods to measure the similarity between real and quantified data distributions, only valid for **"kl_symm_divergence"/"kl_asymm_divergence"** Valid **optimize-algo** values are: 1. **"kl-dis": Default** optimize algorithm, kl divergence 2. **"js-dis":** js divergence, only valid for **"kl_asymm_divergence"** 3. **"cos-dis":** cosine similarity,only valid for **"kl_asymm_divergence** 4. **"mse-dis":** mean square error loss,only valid for **"kl_symm_divergence**
--kl-threshold-size	Whether the feature uses the kl quantization threshold during kl quantization. If the feature size of the corresponding layer > threshold_size, use it. only valid for **"kl_symm_divergence"/"kl-asymm_divergence"。** **2048(default)** The integer must equal 2N where N is a positive integer, recommended values: **2048/4096/8192/12288/16384**
--divergence-nbins	only valid for **"kl_symm_divergence"/"kl-asymm_divergence"** Requires an integer to specify the number of bins in the Kullback-Laiber divergence (KL divergence) histogram. **2048(default)** The integer must equal 2N where N is a positive integer,Recommended value: **2048/4096/8192/12288/16384**
--percentile	Only valid for **"percentile", 0.9999(default)** Suggested value range: **0.999 ~ 0.9999,** step_size **0.0001**
--asymm-percentile	Only valid for "percentile", **-1.0(default)** Suggested value range: **0.999 ~ 0.9999, step_size 0.0001** When it is the default value, it is a symmetric percentage quantization algorithm. When set to other values, it is an asymmetric percentage quantization algorithm.
--moving-alpha	only valid for **"moving_alpha",0.2(default)** Suggested value range: 0.2 ~ 0.9, step_size 0.1
--neg-scale	Scale factor for asymmetric quantization.only valid for **"kl-asymm_divergence"**,**1.0(default)** Suggested value range: **0.001/0.01/0.1/1/1.2/1.5**
--weights-quantize-algo	The weights quantization algorithm.Valid values are: 1. **"min-max": Default** algorithm 2. **"mle":** The mle quantization algorithm,
--weights-optimize-algo	Quantitative methods to measure the similarity between real and quantified data distributions,

	only valid for **"mle"** Valid **optimize-algo** values are: 1.**"kl-dis": Default** optimize algorithm, kl divergence 2.**"js-dis":** js divergence, 3.**"cos-dis":** cosine similarity
--weights-threshold-size	The weights shape size threshold for mle quantization of weights.**2048(default)** recommended values: **2048/4096/8192/12288/16384**
--weights-neg-scale	Scale factor for weights mle quantization.only valid for **"mle"**,**0.999(default)** Suggested value range: **0.99~1.0**
--export-template-code	Whether to export demo code default **True**
--system-env	Specify the single board system environment. Used with --export-template-code ,just to export compiled scripts in different environments. valid values as below: **"linux"**: default **"android"**

4. 其他工具

4.1 Adla_info_tool 工具

Adla_info 是一个能解析生成的 adla 文件的工具。通过 adla_tool 工具，可以获取到 adla 文件对应的平台信息、compiler 版本信息以及输入输出等信息。

Adla_info tool 集成在工具中的路径：

adla-toolkit-binary-x.x.x.x/bin/adla_plug_in/tools/adla_info_tool/ 下，包含下面几个文件：

```
adla-toolkit-binary-1.6.10.3/bin/adla_plug_in/tools/adla_info_tool/
├── adlainfo_release
├── mobilenet_v1_1.0_224_quant_u8.adla
├── README_release_EN.txt
└── README_release.txt
```

客户可以根据 README_release.txt 中的介绍来使用 adla_tool 工具。一般常用命令：

1、./adlainfo_release -v mobilenet_v1_1.0_224_quant_u8.adla 可获取 adla 文件对应的版本信息和平台信息

```
adla file name : mobilenet_v1_1.0_224_quant_u8.adla

**********************************************
Compile_Version : 0.5.3
Aml Hw Version  : 1.0
**********************************************
```

2、 ./adlainfo_release -io mobilenet_v1_1.0_224_quant_u8.adla，获取 adla 文件对应的模型输入输出信息以及 config 中配置的 AXI_SRAM 大小

```
adla file name : mobilenet_v1_1.0_224_quant_u8.adla

****************************************
    Inputs                    : [88 ]
        Input[0]              : 88
        Dim Count             : 4
        Size of dim[0]        : 1
        Size of dim[1]        : 224
        Size of dim[2]        : 224
        Size of dim[3]        : 3
        type                  : Uint8
        scale                 : [0.007812 ]
        zero_point            : [128 ]
        inputs_memory_size    : 150528
****************************************

****************************************
    Outputs                   : [87 ]
        Output[0]             : 87
        Dim Count             : 2
        Size of dim[0]        : 1
        Size of dim[1]        : 1001
        type                  : Uint8
        scale                 : [0.003906 ]
        zero_point            : [0 ]
        outputs_memory_size   : 1001
****************************************
    config:
      max_outstanding_outputs : 4
      axi_sram_size           : 262144 bytes
****************************************
```

4.2 Quantize optimization_tool

Quantize optimization_tool 工具是基于 Adla_tool 工具添加了一些列 feature/weights 量化算法的基础上开发的一个最优量化参数搜索工具。

Quantize optimization_tool 集成在工具中的路径：

adla-toolkit-binary-x.x.x.x/bin/adla_plug_in/tools/Quantize_optimization_tool/ 下，包含下面几个文件：

```
Quantize_optimization_tool/
├── Quantize_optimization_tool_guide_CN.txt
├── Quantize_optimization_tool_guide.txt
├── Quantize_optimization_tool.py
└── Quantize_optimization_tool_simple_version.py
```

客户可根据该目录下的 Model_quantizie_guide 的介绍来使用该工具。

当前工具是在模型转换过程中，将量化后的 IR 导出成 quant tflite 模型，再通过运行 quant tflite 来确认量化效果的。当前工具中的判断标准是通过统计跟原始模型的单帧输出结果的余弦相似度和欧氏距离来判断选取最优值的。

目前已使用该工具提升了多个客户的模型的量化精度效果。

工具使用建议：

1、如果客户的模型有其他的校验集，建议在 run_tflite 接口里面，将检验代码集成进去，根据客户自己的校验结果搜索最优的量化参数。

2、搜索建议：1.修改脚本中 self.quantize_algos=['min-max'],此时会遍历不同数据集的量化效果。优先选取一个在 min-max 量化下的最优量化参数 iterations。 2.修

改 self.iteration=200 的值为第一步中搜索到的最优值，再设置 self.quantize_algos=['kl_asymm_divergence','kl_symm_divergence','percentile','moving-avg','mle'],进行第二轮搜索。

3、详细使用建议参考 Quantize_optimization_tool_guide.txt 文档

4.3 NNAPI_Optimize_tool

NNAPI_Optimize_tool 是针对 NNAPI runtime 对于部分算子不支持导致模型拆分从而影响运行效率的问题，开发了的 一套 对于引入拆分的算子进行重构并在 nnapi 处理流程进行 pass 优化以提升模型运行效率的算子重构工具。

NNAPI_Optimize_tool 集成在工具中的路径：

adla-toolkit-binary-x.x.x.x/bin/adla_plug_in/tools/tflite_refactorer_tool/ 下，包含下面几个文件：

客户可根据该目录下的 Readme 的介绍来使用该工具。只有在客户需要通过 NNAPI 流程加载运行 tflite 模型时才需要使用该工具。

4.4 NNIDE_Tool

NNIDE（Neural Network Integrated Development Environment）是由 Amlogic 开发的一套基于 nnEngine 引擎的综合性神经网络开发工具。NNIDE 为 Amlogic 的客户提供了一个高效、便捷的工作流程，使客户能够更快速地评估其模型在 Amlogic NN 平台上的可行性和性能表现。

NNIDE_TOOL 集成在工具中的路径：

adla-toolkit-binary-x.x.x.x/bin/adla_plug_in/tools/ NNIDE_tool ，包含下面一些文件:

```
1、The directory structure is as follows
├── demo
│   ├── amlnn                          #Compiled file of nnide_tool
│   ├── dataset                        #Dataset directory
│   │   ├── aml_dataset_crop_inception      #Classification model test data set(490 pictures)
│   │   └── input                      #Single frame test data
│   ├── demo_adla_compare_buf.py       #Comparison script between the accuracy of the model file saved on the PC and the accuracy of the board end
│   ├── demo_adla_jupyter.ipynb
│   ├── demo_adla_run_datasets.py      #Classification model top1/top5 accuracy test script
│   ├── demo_adla_run_multi_io.py      #Multi-input model inference script
│   ├── demo_adla_run_single_io.py     #Single input model inference script
│   ├── model_file                     #Test related model files
│   │   ├── adla                           #Converted adla file
│   │   └── original                       #Original model files, and conversion scripts
│   ├── NNIDE_package                  #Board end IDE executable file
│   │   ├── android_32                 #Android 32-bit system IDE execution file
│   │   │   └── NNIDE
│   │   ├── android_64                 #Android 64-bit system IDE execution file
│   │   │   └── NNIDE
│   │   ├── linux_32                   #Linux 64-bit system IDE execution file
│   │   │   └── NNIDE
│   │   └── yocto_64                   #Yocto 64-bit system IDE execution file
│   │   └── NNIDE
│   └── requirements.txt               #PC environment dependency list
└── Readme.txt

Note:
    1. Unzip command: tar -zxf demo.tar.gz
```

客户基于该工具，可以直接在 PC 端 python 环境下获取到模型在板端运行结果，并直接使用 python 实现模型前后处理，可以快速测试模型在板端运行的准确性。

具体参考：NNIDE_1.x.x_CN.docx